CHAIN HOMOTOPIES BETWEEN N(A) AND N(B) LIFT TO SIMPLICIAL HOMOTOPIES U X DELTA[1] -> V

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sources/stacks-simplicial/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the elementwise computation of delta(sigma) is announced as done 'in case A is the category of abelian groups' (source line 5586) - the standard reduction of morphisms between direct sums to their component maps, which is legitimate because both sides are additive functors in U and the constructed morphisms are natural; every cancellation in the computation is written out and the surviving term is identified; this lemma is the converse companion of the homotopy correspondence of section-homotopy-abelian.

Editorial note. The label lemma-backwards-homotopy names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- The diamond functor representing homotopy data. Same chapter, \label{lemma-represent-homotopy}. Lines 5522-5526: the triple (N(a), N(b), {N_n}) is represented by a morphism diamond N(U) -> N(V); reused at line 5573 for Psi.
- The explicit s-level homotopy between e_0 and e_1. Same chapter, \label{lemma-homotopy-s-N}. Lines 5547-5552: the explicit homotopy {h_n : s(U)_n -> s(U x Delta[1])_{n+1}} between s(e_0) and s(e_1).
- The map-into-diamond factorization criterion. Same chapter, \label{lemma-map-into-diamond}. Lines 5553-5556: produces Phi : diamond s(U) -> s(U x Delta[1]) from the homotopy; lines 5576-5581: produces Psi from the factored delta(sigma).
- Simplicial homotopies and the cylinder object. Same chapter, \label{definition-homotopy} and \label{equation-homotopy-to-homotopy}. Lines 5531-5535 and 5545-5548: e_0, e_1 : U -> U x Delta[1] are homotopic with homotopy id; N(h) conventions from section-homotopy-abelian.
- Termwise split short exact sequences and the connecting morphism. Homology chapter, \label{homology-lemma-ses-termwise-split}. Lines 5584-5588: delta(sigma) is computed as d sigma - sigma d.

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- N is a direct summand of s. Structural property of the normalized complex functor (Same chapter, Section \ref{section-homotopy-abelian}). Lines 5537-5541: the reduction from N to s uses that N is a direct summand of s and that diamond commutes with direct sums.

Source reference. Original reference 019K: sources/stacks-simplicial/pinned-original-source.tex, source lines 4640–4640. Prerequisite pointer invoked at proof line(s) 5515; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original equation 019R: sources/stacks-simplicial/pinned-original-source.tex, source lines 4971–4971. Prerequisite pointer invoked at proof line(s) 5532; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 011D: sources/stacks-chow/tags.txt, source lines 1359–1359. Prerequisite pointer invoked at proof line(s) 5607; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 019S: sources/stacks-simplicial/pinned-original-source.tex, source lines 5034–5034. Prerequisite pointer invoked at proof line(s) 5533; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 01A3: sources/stacks-simplicial/pinned-original-source.tex, source lines 5434–5434. Prerequisite pointer invoked at proof line(s) 5559; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 01A2: sources/stacks-simplicial/pinned-original-source.tex, source lines 5335–5335. Prerequisite pointer invoked at proof line(s) 5523; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original section 019J: sources/stacks-simplicial/pinned-original-source.tex, source lines 4614–4614. Prerequisite pointer invoked at proof line(s) 5580; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original section 019Q: sources/stacks-simplicial/pinned-original-source.tex, source lines 4955–4955. Prerequisite pointer invoked at proof line(s) 5518; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let $\mathcal{A}$ be an abelian category. Let U, V be simplicial objects of $\mathcal{A}$. Let $a, b : U \to V$ be a pair of morphisms. Assume the corresponding maps of chain complexes $N(a), N(b) : N(U) \to N(V)$ are homotopic by a homotopy $\{N_n : N(U)_n \to N(V)_{n+1}\}$. Then there exists a homotopy from a to b as in Definition 019K. Moreover, one can choose the homotopy $h : U \times \Delta[1] \to V$ such that $N_n = N(h)_n$ where $N(h)$ is the homotopy coming from h as in Section 019Q.*

Proof. Let $(\diamond N(U), \diamond a, \diamond b, \diamond h)$ be as in Lemma 01A2 and its proof. By that lemma there exists a morphism $\diamond N(U) \to N(V)$ representing the triple $(N(a), N(b), \{N_n\})$. We will show there exists a morphism $\psi : N(U \times \Delta[1]) \to \diamond N(U)$ such that $\diamond a = \psi \circ N(e_0)$, and $\diamond b = \psi \circ N(e_1)$. Moreover, we will show that the homotopy between $N(e_0), N(e_1) : N(U) \to N(U \times \Delta[1])$ coming from (019R) and Lemma 019S with $h = \mathrm{id}_{U \times \Delta[1]}$ is mapped via ψ to the canonical homotopy $\diamond h$ between the two maps $\diamond a, \diamond b : N(U) \to \diamond N(U)$. Certainly this will imply the lemma.

Note that $N : \mathrm{Simp}(\mathcal{A}) \to \mathrm{Ch}_{\geq 0}(\mathcal{A})$ as a functor is a direct summand of the functor $s : \mathrm{Simp}(\mathcal{A}) \to \mathrm{Ch}_{\geq 0}(\mathcal{A})$. Also, the functor $\diamond$ is compatible with direct sums. Thus it suffices instead to construct a morphism $\Psi : s(U \times \Delta[1]) \to \diamond s(U)$ with the corresponding properties. This is what we do below.

By Definition 019K the morphisms $e_0 : U \to U \times \Delta[1]$ and $e_1 : U \to U \times \Delta[1]$ are homotopic with homotopy $\mathrm{id}_{U \times \Delta[1]}$. By Lemma 019S we get an explicit homotopy $\{h_n : s(U)_n \to s(U \times \Delta[1])_{n+1}\}$ between the morphisms of chain complexes $s(e_0) : s(U) \to s(U \times \Delta[1])$ and $s(e_1) : s(U) \to s(U \times \Delta[1])$. By Lemma 01A3 above we get a corresponding morphism

$$\Phi : \diamond s(U) \to s(U \times \Delta[1])$$

According to the construction, Φ_n restricted to the summand $s(U)[-1]_n = s(U)_{n-1}$ of $\diamond s(U)_n$ is equal to h_{n-1}. And

$$h_{n-1} = \sum\nolimits_{i=0}^{n-1} (-1)^{i+1} s_i^n \cdot \alpha_{i+1}^n : U_{n-1} \to \bigoplus\nolimits_j U_n \cdot \alpha_j^n.$$

with obvious notation.

On the other hand, the morphisms $e_i : U \to U \times \Delta[1]$ induce a morphism $(e_0, e_1) : U \oplus U \to U \times \Delta[1]$. Denote W the cokernel. Note that, if we write $(U \times \Delta[1])_n = \bigoplus_{\alpha : [n] \to [1]} U_n \cdot \alpha$, then we may identify $W_n = \bigoplus_{i=1}^n U_n \cdot \alpha_i^n$ with α_i^n as in Section 019J. We have a commutative diagram

$$\begin{array}{ccccccccc} 0 & \longrightarrow & U \oplus U & \longrightarrow & U \times \Delta[1] & \longrightarrow & W & \longrightarrow & 0 \\ & & & \searrow^{(1,1)} & \downarrow \pi & & & & \\ & & & & U & & & & \end{array}$$

This implies we have a similar commutative diagram after applying the functor s. Next, we choose the splittings $\sigma_n : s(W)_n \to s(U \times \Delta[1])_n$ by mapping the summand $U_n \cdot \alpha_i^n \subset W_n$ via $(-1, 1)$ to the summands $U_n \cdot \alpha_0^n \oplus U_n \cdot \alpha_i^n \subset (U \times \Delta[1])_n$. Note that $s(\pi)_n \circ \sigma_n = 0$. It follows that $(1, 1) \circ \delta(\sigma)_n = 0$. Hence $\delta(\sigma)$ factors as in Lemma 01A3. By that lemma we obtain a canonical morphism $\Psi : s(U \times \Delta[1]) \to \diamond s(U)$.

To compute Ψ we first compute the morphism $\delta(\sigma) : s(W) \to s(U)[-1] \oplus s(U)[-1]$. According to Homology, Lemma 011D and its proof, to do this we have compute

$$d_{s(U \times \delta[1]), n} \circ \sigma_n - \sigma_{n-1} \circ d_{s(W), n}$$

and write it as a morphism into $U_{n-1} \cdot \alpha_0^{n-1} \oplus U_{n-1} \cdot \alpha_n^{n-1}$. We only do this in case $\mathcal{A}$ is the category of abelian groups. We use the short hand notation x_α for $x \in U_n$ to denote the element x in the summand $U_n \cdot \alpha$ of $(U \times \Delta[1])_n$. Recall that

$$d_{s(U\times\delta[1]),n} = \sum\nolimits_{i=0}^{n} (-1)^i d_i^n$$

where d_i^n maps the summand $U_n \cdot \alpha$ to the summand $U_{n-1} \cdot (\alpha \circ \delta_i^n)$ via the morphism d_i^n of the simplicial object U. In terms of the notation above this means

$$d_{s(U\times\delta[1]),n}(x_\alpha) = \sum\nolimits_{i=0}^{n} (-1)^i (d_i^n(x))_{\alpha\circ\delta_i^n}$$

Starting with $x_\alpha \in W_n$, in other words $\alpha = \alpha_j^n$ for some $j \in \{1, \ldots, n\}$, we see that $\sigma_n(x_\alpha) = x_\alpha - x_{\alpha_0^n}$ and hence

$$(d_{s(U\times\delta[1]),n} \circ \sigma_n)(x_\alpha) = \sum\nolimits_{i=0}^{n} (-1)^i (d_i^n(x))_{\alpha\circ\delta_i^n} - \sum\nolimits_{i=0}^{n} (-1)^i (d_i^n(x))_{\alpha_0^n\circ\delta_i^n}$$

To compute $d_{s(W),n}(x_\alpha)$, we have to omit all terms where $\alpha \circ \delta_i^n = \alpha_0^{n-1}, \alpha_n^{n-1}$. Hence we get

$$(\sigma_{n-1} \circ d_{s(W),n})(x_\alpha) =$$
$$\sum\nolimits_{i=0,\ldots,n \text{ and } \alpha\circ\delta_i^n \neq \alpha_0^{n-1} \text{ or } \alpha_n^{n-1}} \left((-1)^i (d_i^n(x))_{\alpha\circ\delta_i^n} - (-1)^i (d_i^n(x))_{\alpha_0^{n-1}} \right)$$

Clearly the difference of the two terms is the sum

$$\sum\nolimits_{i=0,\ldots,n \text{ and } \alpha\circ\delta_i^n = \alpha_0^{n-1} \text{ or } \alpha_n^{n-1}} \left((-1)^i (d_i^n(x))_{\alpha\circ\delta_i^n} - (-1)^i (d_i^n(x))_{\alpha_0^{n-1}} \right)$$

Of course, if $\alpha \circ \delta_i^n = \alpha_0^{n-1}$ then the term drops out. Recall that $\alpha = \alpha_j^n$ for some $j \in \{1, \ldots, n\}$. The only way $\alpha_j^n \circ \delta_i^n = \alpha_n^{n-1}$ is if $j = n$ and $i = n$. Thus we actually get 0 unless $j = n$ and in that case we get $(-1)^n (d_n^n(x))_{\alpha_n^{n-1}} - (-1)^n (d_n^n(x))_{\alpha_0^{n-1}}$. In other words, we conclude the morphism

$$\delta(\sigma)_n : W_n \to (s(U)[-1] \oplus s(U)[-1])_n = U_{n-1} \oplus U_{n-1}$$

is zero on all summands except $U_n \cdot \alpha_n^n$ and on that summand it is equal to $((-1)^n d_n^n, -(-1)^n d_n^n)$. (Namely, the first summand of the two corresponds to the factor with α_n^{n-1} because that is the map $[n-1] \to [1]$ which maps everybody to 0, and hence corresponds to e_0.)

We obtain a canonical diagram

$$\begin{array}{ccccccccc}
0 & \longrightarrow & s(U) \oplus s(U) & \longrightarrow & \diamond s(U) & \longrightarrow & s(U)[-1] & \longrightarrow & 0 \\
 & & \downarrow & & \downarrow{\scriptstyle \Phi} & & \downarrow & & \\
0 & \longrightarrow & s(U) \oplus s(U) & \longrightarrow & s(U \times \Delta[1]) & \longrightarrow & s(W) & \longrightarrow & 0 \\
 & & \downarrow & & \downarrow{\scriptstyle \Psi} & & \downarrow & & \\
0 & \longrightarrow & s(U) \oplus s(U) & \longrightarrow & \diamond s(U) & \longrightarrow & s(U)[-1] & \longrightarrow & 0
\end{array}$$

We claim that $\Phi \circ \Psi$ is the identity. To see this it is enough to prove that the composition of Φ and $\delta(\sigma)$ as a map $s(U)[-1] \to s(W) \to s(U)[-1] \oplus s(U)[-1]$ is the identity in the first factor and minus identity in the second. By the computations above it is $((-1)^n d_0^n, -(-1)^n d_0^n) \circ (-1)^n s_n^n = (1, -1)$ as desired. $\square$